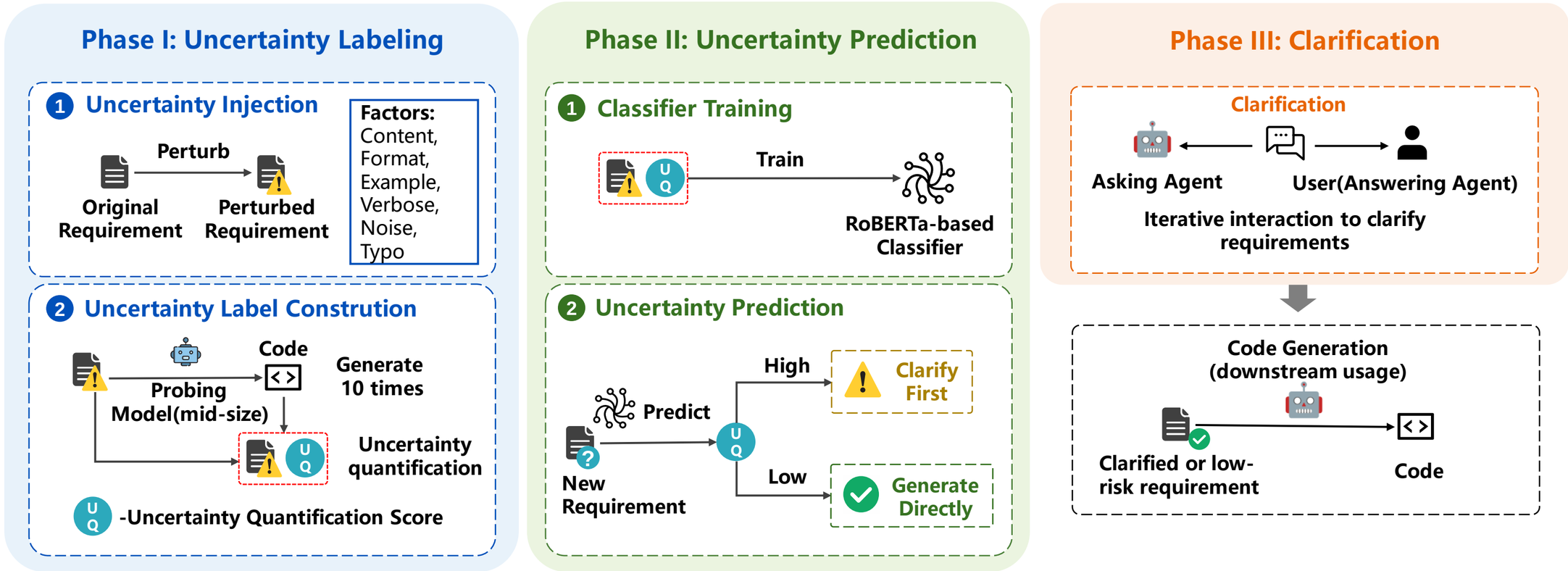

Phase I: Uncertainty Labeling
1 Uncertainty Injection
Perturb
Original Requirement
Perturbed Requirement
Factors: Content, Format, Example, Verbose, Noise, Typo
2 Uncertainty Label Constrution
Code
Probing Model(mid-size)
Generate 10 times
Uncertainty quantification
UQ -Uncertainty Quantification Score
Phase II: Uncertainty Prediction
1 Classifier Training
Train
RoBERTa-based Classifier
2 Uncertainty Prediction
Predict
New Requirement
High
Clarify First
Low
Generate Directly
Phase III: Clarification
Clarification
Asking Agent
User(Answering Agent)
Iterative interaction to clarify requirements
Code Generation (downstream usage)
Clarified or low-risk requirement
Code